

Article No. : 1FK2206-2AF01-0SB0-Z
C32+M01+R20



Figure similar

Client order no. :
Order no. :
Offer no. :
Remarks :

Item no. :
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Project :

Basic data of geared motor	
Motor type	Permanent-magnet synchronous motor, Planetary gearbox, Natural cooling, Degree of protection IP65
Motor type	Compact
Static torque at output $M_{2,0}$	100.00 Nm
Static current I_0	4.2 A
Maximum torque at output $M_{2max}^{1)}$	192.00 Nm
Maximum output speed n_{2max}	350 rpm
Moment of inertia motor + gearbox (related to the input) J_1	9.082 kgcm ²
Mass m	10.90 kg
Lubrication	Standard

Rated data of geared motor	
SINAMICS S120, ALM 3AC 400V	
Rated speed related to the gear output n_{2N}	100 rpm
Rated torque related to the gear output M_{2N}	62.00 Nm
Rated power P_N	0.649 kW

Basic data of gearbox	
Gearbox type and size	Planetary gearbox NLC080
Transmission ratio i	1 : 20 (Output to input)
Number of gear stages z	2
Output torque (fatigue strength) $M_{2N,G}$	120.0 Nm
Maximum permissible output torque (short-time, end of fatigue strength) $M_{2max,G}^{2)}$	192.0 Nm
Emergency off output moment (1000 cycles) $M_{2Em,Off}$	240.0 Nm
Torsional backlash related to the output φ_2	9 '
Torsional stiffness related to the output c_{T2}	11.8 Nm/'
Maximum static radial force F_{Rmax}	5,500 N
Max. average radial force for 20000 h $F_{Req}^{3)}$	5,500 N
Maximum static axial force F_{Amax}	6,400 N
Max. average axial force for 20000 h $F_{Aeq}^{4)}$	6,400 N
Max. average breakdown torque M_K	Nm
Max. bending moment on the flange to the motor M_B	16 Nm
Efficiency η_G	0.94
Degree of protection gearbox	IP65
Gearbox shaft end	Fitted key

Basic motor data	
Maximum average torque (incl. derating due to mounted gearing) $M_{0,M}$	5.04 Nm
Maximum average continuous current (incl. derating due to mounted gearing) $I_{0,M}$	3.26 A
Maximum acceleration torque $M_{max,M}^{2)}$	17.30 Nm
Maximum short-time permissible current $I_{max,M}$	17.80 A
Degree of protection motor	IP65
Connection type	Double cable connection for S120
Connector size	M23
Encoder system	Encoder AS22DQC: Absolute encoder single turn 22 bit
Color of the housing	Standard (Anthracite, similar to RAL 7016)

¹⁾Fatigue limit range - for max. 30 000 revolutions of the output shaft, utilization only with service life calculation

²⁾The maximum acceleration torque $M_{max,M}$ x of transmission ratio i is greater than the maximum permitted output torque (short-time fixed) $M_{2max,G}$. Depending on the load conditions, a torque limitation and service life calculation may be necessary.

³⁾based on an output speed of 100 rpm and a force application point in the center of the shaft

⁴⁾based on an output speed of 100 rpm